

JPEG Recompression Residuals for Image Forgery Screening

A concise technical synthesis of compression-history cues

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ABSTRACT	This paper summarizes how unequal JPEG compression histories can produce local residual differences in manipulated images. It outlines a transparent screening workflow based on controlled recompression, pixel residuals, and block-level comparison. The discussion distinguishes an anomaly cue from proof of tampering and proposes evaluation conditions needed for a future empirical study. No new experiments or performance results are claimed.
KEYWORDS	JPEG forensics; double compression; DCT; recompression residual; copy-move; splicing

1. Introduction

Digital images are routinely used as evidence and as a means of communication, while copy-move and splicing operations can leave changes that are difficult to see. Passive forensic methods therefore examine traces created by image processing rather than relying only on visual content. This short paper synthesizes the compression-based rationale in the source thesis and frames it as an image-level screening problem [1, 2].

The central question is whether local regions respond differently to a controlled JPEG recompression. The paper is a theoretical synthesis: it proposes a transparent analysis workflow, but reports no new experiments or performance claims.

2. JPEG compression history as a forensic cue

JPEG transforms non-overlapping 8 x 8 pixel blocks into frequency coefficients and quantizes those coefficients. Quantization is lossy; repeated encoding can therefore leave statistical and pixel-level traces. If image regions have different processing histories, a later uniform save may not erase every local discrepancy. Spliced content, for example, can carry a compression history or block-grid alignment that differs from the surrounding image. Copy-move operations may also disturb local traces when the copied area is transformed or the composite is re-encoded [1, 2].

This mechanism is a cue, not a rule that every manipulated region must be singly compressed while every background region is doubly compressed. The observed response depends on the source files, quantization tables, edits, alignment, and subsequent platform processing. A detector should therefore estimate inconsistency and report uncertainty rather than treat a single residual pattern as proof.

3. A compact screening workflow

A reproducible screen can be organized in four steps. First, retain the original JPEG and record its dimensions and available encoding metadata. Second, decode and recompress a working copy at one or more declared quality settings. Third, calculate a pixel residual and summarize it locally. Fourth, compare each block with its spatial neighborhood and flag outlying areas for review.

$$D_Q(x,y) = | I(x,y) - J_Q(I)(x,y) |$$

Here, I is the decoded input and $J_Q(I)$ is its decode-after-recompression at quality setting Q . A block summary may include the mean and spread of D_Q , along with features from quantized DCT coefficients. Using multiple Q settings can show whether a suspected anomaly persists, but the settings and any intensity scaling must be disclosed. The output should separate image-level screening (whether an image is suspicious) from localization (where an anomaly occurs).

A learned classifier may use the residual map or block statistics as features. Recompression-based CNN systems in the literature illustrate this direction [1, 2]. A lightweight baseline should still be retained so that decisions can be checked against interpretable residual maps.

4. Evaluation and limitations

A future validation study should include authentic images and controlled copy-move and splicing edits, with source-image-disjoint train, validation, and test partitions. It should vary the initial JPEG quality, final save quality, block-grid alignment, image resizing, and social-media recompression. Image-level precision, recall, F1, and ROC-AUC are suitable for screening; pixel-level intersection-over-union or Dice scores are needed only when localization masks are evaluated.

Texture, edges, camera pipelines, screenshots, resizing, and repeated benign saves can all create residual variation. Conversely, careful editing or strong later compression can weaken traces. Quality-factor selection and residual amplification can also change the map. Thus, compression evidence should be corroborated with other forensic cues or provenance information. No conclusion about authorship, intent, or authenticity should be drawn from ELA or JPEG residuals alone.

5. Conclusion

JPEG recompression can expose local differences in compression history and offers a practical cue for forgery screening. A defensible system preserves the source, declares its recompression settings, summarizes residuals at block level, and communicates uncertainty. Benchmarking across editing and distribution conditions is required before any detector can be described as robust.

References

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- [3] N. T. Lam, *Deep Learning Methods for Image Forgery Detection*, undergraduate thesis, Can Tho University, 2026. Chapters 1–2: research overview and theoretical background.

Source note. This concise conceptual paper was prepared from Chapters 1–2 (research overview and theoretical background) of the undergraduate thesis *Deep Learning Methods for Image Forgery Detection* (Nguyen Thuy Lam, 2026). It introduces no new dataset, experiment, or measured performance.